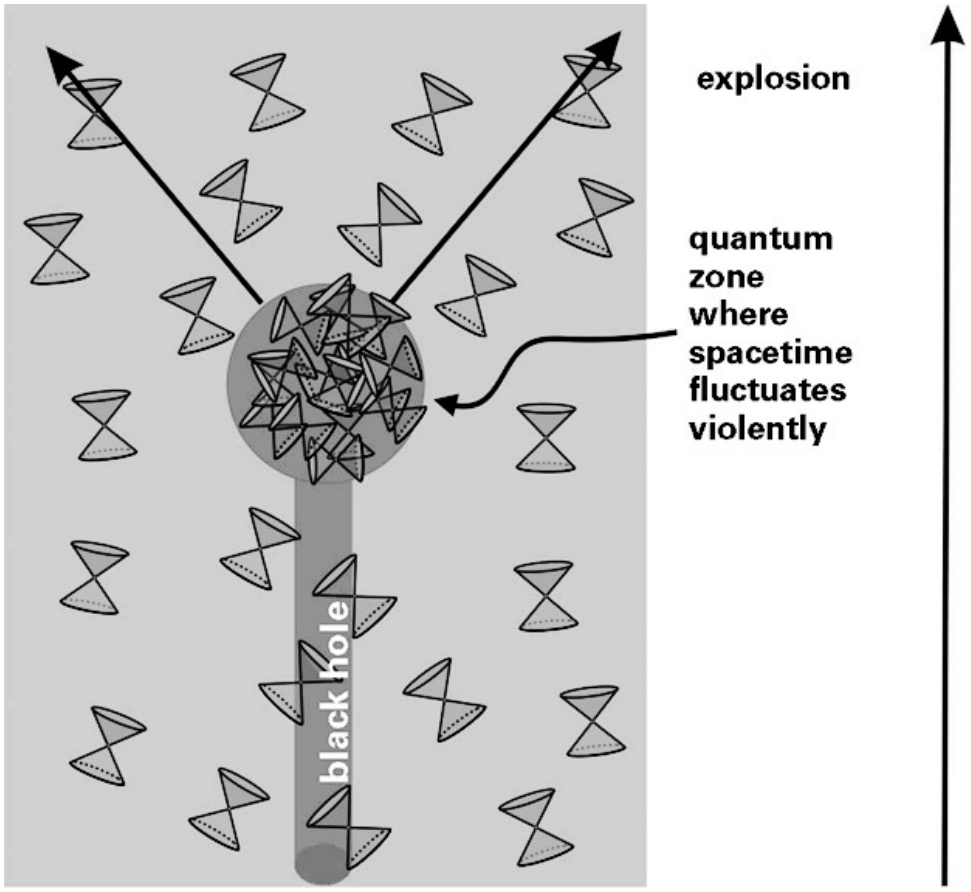


During the course of this phase, inside the black hole and within its immediate vicinity there is no longer a single and determinate spacetime. There is a quantum superposition of spin webs. Just as an electron can unfurl into a cloud of probabilities between the moment it is emitted and the moment it arrives on a screen, passing through more than one place, so the spacetime of the quantum collapse of a black hole passes through a phase in which time fluctuates violently, there is a quantum superposition of different times, and then, later, a return to a determined state after the explosion.

For this intermediate phase, where time is wholly indeterminate, we still have equations that tell us what happens. Equations without time.

This is the world described by loop theory.

Am I certain that this is the correct description of the world? I am not, but it is today the only coherent and complete way that I know of to think about the structure of spacetime without neglecting its quantum properties. Loop quantum gravity shows that it is possible to write a coherent theory without fundamental space and time—and that it can be used to make qualitative predictions.

In a theory of this kind, time and space are no longer containers or general forms of the world. They are approximations of a quantum dynamic that in